EduFin SQL Business Challenge

Phase 1 — Monday Submission

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STEP 1A — Understand the Data Structure

```
SELECT
    COUNT(*) as total_loans,
    COUNT(DISTINCT loan_id) as unique_loan_ids,
    COUNT(*) - COUNT(loan_status) AS null_loan_status_count
FROM loans;

-- There are total 5000 rows
-- loan_id is the primary_key
-- There are no null values in loan_status
```

```
+-----------+---------------+----------------------+
|total_loans|unique_loan_ids|null_loan_status_count|
+-----------+---------------+----------------------+
|       5000|           5000|                     0|
+-----------+---------------+----------------------+
```

STEP 1B — Basic Portfolio Metrics

```
SELECT
    COUNT(*) as total_loans,
    ROUND(SUM(loan_amount) / 10000000.0, 2) AS total_disbursed,
    COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) AS total_deafulted
FROM loans;

-- Total loans: 5000
-- Total amount disbursed (in crores): 204.82
-- Number of deafaulted loans: 590
```

```
+-----------+-----------------------------+---------------+
|total_loans|total_amount_disbursed_crores|defaulted_loans|
+-----------+-----------------------------+---------------+
|       5000|                       204.82|            590|
+-----------+-----------------------------+---------------+
```

STEP 1C — Calculate Risk Metrics

```
SELECT
    ROUND(
        COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*),
        2
    ) AS default_rate,
    ROUND(
        (SUM(CASE WHEN loan_status = 'Defaulted' THEN loan_amount END) /
SUM(loan_amount)) * 100,
        2
    ) AS loss_rate
FROM loans

-- EduFin is ABOVE the 10% threshold
-- The classification triggers High Risk.
```

default_rate	loss_rate
11.80	11.83

STEP 1D — Financial Impact (Crores Format)

```
SELECT
    ROUND(SUM(loan_amount) / 10000000.0, 2) AS defaulted_amount_crores
FROM loans
WHERE loan_status = 'Defaulted';

-- The average loan size is approximately 0.041 Cr (204.82 Cr / 5000 loans). With 590
defaulted loans, the estimated loss is ~24.16 Cr. This closely matches the actual
defaulted amount of 24.24 Cr, confirming the calculations are consistent.
```

defaulted_amount_crores
24.24

STEP 1E — Executive Dashboard (Final Consolidated Query)

```
SELECT
    COUNT(*) AS total_loans,
    COUNT(CASE WHEN loan_status = 'Active' THEN 1 END) AS active_loans,
    COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) AS defaulted_loans,
    COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END) AS overdue_loans,
    COUNT(CASE WHEN loan_status = 'Closed' THEN 1 END) AS closed_loans,
    ROUND(
        COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 / COUNT(*),
        2
    ) AS default_rate_pct,
    ROUND(
        SUM(CASE WHEN loan_status = 'Defaulted' THEN loan_amount END) / 10000000.0,
        2
    ) AS total_loss_cr,
    CASE
```

```
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) <= 5.00 THEN 'HEALTHY'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) <= 10.00 THEN 'MODERATE RISK'
        WHEN ROUND(COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) * 100.0 /
COUNT(*), 2) <= 15.00 THEN 'HIGH RISK'
        ELSE 'CRITICAL'
    END AS portfolio_health_status
FROM loans
```

total_loans	active_loans	defaulted_loans	overdue_loans	closed_loans	default_rate_pct	total_loss_cr	portfolio_health_status
5000	3965	590	273	172	11.80	24.24	HIGH RISK

Custom Column

```
CASE
    WHEN (COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) +
            COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END)) * 100.0 / COUNT(*) <
5.00
        THEN 'Stable'
    WHEN (COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) +
            COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END)) * 100.0 / COUNT(*) <
8.00
        THEN 'Monitor'
    WHEN (COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) +
            COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END)) * 100.0 / COUNT(*) <
12.00
        THEN 'Alert'
    WHEN (COUNT(CASE WHEN loan_status = 'Defaulted' THEN 1 END) +
            COUNT(CASE WHEN loan_status = 'Overdue' THEN 1 END)) * 100.0 / COUNT(*) <
16.00
        THEN 'Urgent'
    ELSE 'Emergency'
END AS risk_tier
```

> I designed EduFin's custom threshold to count both defaulted (590) and overdue (273) loans together as "effective default rate" because overdue loans are defaults-waiting-to-happen—they will become defaults within 6-12 months. This changes the picture from 11.80% (looks manageable) to 17.26% (clearly TIER 4: Crisis), forcing honest board-level action instead of complacency. Standard thresholds hide this future risk by only counting existing defaults; my threshold makes it visible today.

WHY Gate

1. Does your "Financial Impact" (Step 1D) include only defaulted loans, or does it include "Overdue" as well? Why did you make that choice?

> My Financial Impact (Step 1D) includes ONLY defaulted loans (₹24.24 Cr), NOT overdue loans. This is because defaulted loans are REALIZED losses (money already gone, unrecoverable). Overdue loans are POTENTIAL losses (still in collections process, recovery possible). The board

needs to distinguish between these two: • defaulted_amount = Actual capital lost (this quarter's damage) • overdue_amount = At-risk but potentially recoverable (next quarter's risk) Mixing them would overstate current losses and blur the true financial position.

2. If the CEO asks if this crisis is "temporary," what specific data point from your analysis would you point to as a warning sign that it's systemic?

> The 3.44% success rate (only 172 loans repaid out of 5,000) proves this is systemic, not temporary. Nearly 97% of our portfolio is stuck in default or overdue status across all loan cohorts. If the problem was temporary (bad market, bad luck), old loans would be fine by now. They're not. Almost all our loans are failing both old ones and new ones. This means we're picking the wrong borrowers from the start. It's not a temporary problem. It's a hiring problem in our lending team.

3. What is the single most important number in this workbook that the board should see first? Why that one?

> ₹24.24 Crores Because it's the ONLY number that matters to the board, actual capital lost. Everything else (default rate %, loan counts, risk tiers) is just explanation for why we lost ₹24.24 Cr. The board doesn't care about percentages or classifications. They care about: "How much money is gone?"

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